

SAN ANGELO, TX, USA

WMO: 722630

Lat: 31.352N

Lon: 100.495W

Elev: 1916

StdP: 13.71

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 23034

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	22.1	25.8	6.1	8.1	38.1	10.4	10.1	40.4	28.6	61.8	26.0	56.1	7.4	30	0.508

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.6	102.2	70.3	100.0	70.2	97.9	70.2	75.4	90.1	74.4	89.0	73.5	88.0	10.5	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
71.8	126.2	80.1	70.8	121.8	79.3	69.9	117.9	78.8	40.2	89.8	39.2	88.7	38.4	88.1	80.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.4	20.9	19.0	15.6	105.6	3.6	2.8	13.0	107.6	10.9	109.3	9.0	110.9	6.4	113.0
			13.9	77.1	3.5	1.6	11.4	78.2	9.3	79.1	7.3	80.0	4.8	81.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	66.6	47.4	51.5	59.1	67.1	75.3	81.8	84.2	83.8	76.8	67.0	56.1	48.2	(6)
	DBStd	15.47	9.52	10.35	9.71	8.49	7.31	5.10	4.16	4.55	6.22	8.11	9.11	9.15	(7)
	HDD50	488	162	97	32	4	0	0	0	0	0	3	44	146	(8)
	HDD65	2138	546	387	231	75	11	0	0	0	4	76	286	522	(9)
	CDD50	6555	82	138	313	517	786	955	1062	1047	805	530	229	91	(10)
	CDD65	2731	2	8	47	139	332	505	597	582	360	137	20	2	(11)
	CDH74	33171	61	181	575	1690	3779	6215	7789	7357	3853	1391	224	56	(12)
	CDH80	17458	8	46	172	697	1896	3416	4522	4246	1904	504	39	8	(13)
(14) Wind	WSAvg	9.3	9.6	10.1	10.6	11.2	10.3	9.8	8.6	7.7	7.6	8.4	8.9	9.1	(14)
(15) Precipitation	PrecAvg	20.9	0.9	1.1	1.2	1.6	3.0	2.3	1.1	2.2	3.0	2.5	1.1	0.9	(15)
	PrecMax	35.8	3.7	4.5	4.6	5.1	11.3	7.0	4.6	8.1	11.0	11.4	5.2	4.0	(16)
	PrecMin	9.8	0.0	0.0	0.0	0.0	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	(17)
	PrecStd	6.3	0.9	1.1	1.0	1.3	2.2	1.7	1.0	1.9	2.3	2.2	1.1	0.9	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	80.5	86.0	89.7	96.6	101.7	105.0	104.2	104.6	101.2	93.2	84.7	80.1	(19)
		MCWB	55.4	57.2	60.1	63.0	67.3	70.6	70.4	71.1	69.3	67.7	62.4	56.8	(20)
	2%	DB	74.5	79.3	84.8	91.8	97.5	101.0	101.4	101.7	96.6	88.8	79.8	74.3	(21)
		MCWB	53.8	56.7	59.8	63.1	67.6	70.2	70.4	71.2	69.7	67.4	61.1	56.0	(22)
	5%	DB	69.4	74.6	80.5	87.7	94.2	97.8	99.2	99.3	93.5	85.9	76.0	69.4	(23)
		MCWB	52.7	55.5	59.2	63.1	67.4	70.3	70.6	70.8	69.4	65.9	60.3	54.2	(24)
	10%	DB	64.8	69.7	76.1	83.7	90.3	94.7	97.1	96.7	90.5	82.2	72.1	64.9	(25)
		MCWB	51.1	54.0	58.3	62.3	67.8	70.9	70.6	70.7	69.3	64.7	58.8	52.7	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	62.2	64.2	66.6	70.7	75.2	77.2	76.4	76.0	75.2	73.3	68.1	64.0	(27)
		MCDB	69.2	73.7	78.1	83.7	89.6	92.0	91.6	91.6	88.3	83.8	76.2	70.4	(28)
	2%	WB	59.0	62.0	64.9	68.9	73.4	75.5	74.9	74.6	73.5	71.2	65.7	60.4	(29)
		MCDB	67.0	71.0	75.6	80.9	86.6	89.4	90.1	90.6	86.0	81.8	73.0	67.5	(30)
	5%	WB	55.8	59.5	63.3	67.6	72.1	74.3	73.8	73.7	72.6	69.6	63.5	57.2	(31)
		MCDB	65.0	68.6	73.7	78.9	84.9	88.3	89.0	89.3	84.9	79.7	71.8	66.0	(32)
	10%	WB	52.6	56.6	61.5	66.1	70.8	73.2	72.9	72.9	71.5	68.1	61.2	53.8	(33)
		MCDB	63.2	67.2	71.8	77.4	83.7	87.2	88.1	88.2	83.5	77.2	69.7	63.3	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	26.0	26.0	25.8	27.0	24.4	23.0	23.6	23.2	23.2	24.6	24.7	25.3	(35)
		MCDBR	33.5	33.3	30.8	31.1	29.2	27.1	26.1	26.1	27.0	28.4	28.9	32.2	(36)
	5% WB	MCWBR	17.8	17.2	14.4	13.4	10.5	7.9	7.4	7.5	9.0	11.8	14.6	17.2	(37)
		MCWBR	26.3	26.2	23.9	24.3	23.8	22.5	22.7	22.8	21.6	21.0	21.4	24.5	(38)
(40) Clear-Sky Solar Irradiance	taub	taub	0.286	0.294	0.317	0.354	0.374	0.390	0.401	0.389	0.373	0.336	0.308	0.291	(40)
		taud	2.564	2.545	2.492	2.399	2.406	2.400	2.425	2.453	2.426	2.522	2.529	2.552	(41)
	Ebn at Noon	Ebn at Noon	302	309	307	297	290	283	280	283	285	290	291	294	(42)
		Edh at Noon	26	29	33	38	38	38	37	35	35	30	27	25	(43)
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1047	1278	1646	1989	2069	2263	2203	2033	1690	1375	1099	938	(44)
	RadStd	RadStd	110	166	124	150	180	156	148	120	122	197	134	109	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	+0.68	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+230	+157	(46)
(47)	Regional (19 neighbors)	+0.72	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+252	+180	(47)

Nomenclature: See separate page